ELEC-337
ELECTRONICS LABORATORY-II

EXPERIMENT 4

Basic Output Stage Topologies

Name:.....

Surname: .....

Student ID:.....

Section:.....

Date:.....

OBJECTIVE: *Studying the characteristics of different output stage topologies*



DEPARTMENT OF ELECTRONICS ENGINEERING
GEBZE TECHNICAL UNIVERSITY

This experiment has been adopted from Department of Electrical and Electronics Engineering, Boğaziçi University

PROCEDURE

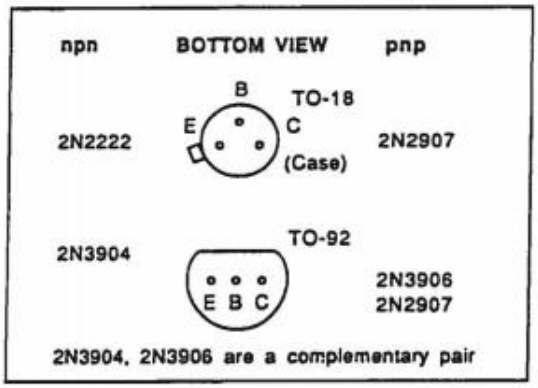


Figure 1: 2N2222 npn and 2N2907 pnp BJT pin configurations

Be sure to include power-supply bypass capacitors (100nF) on each supply. You will use 2N222 npn , 2N2907 pnp transistors and 1N4148 diodes throughout the experiment.

1. Class A Follower

1.1 DC Bias:

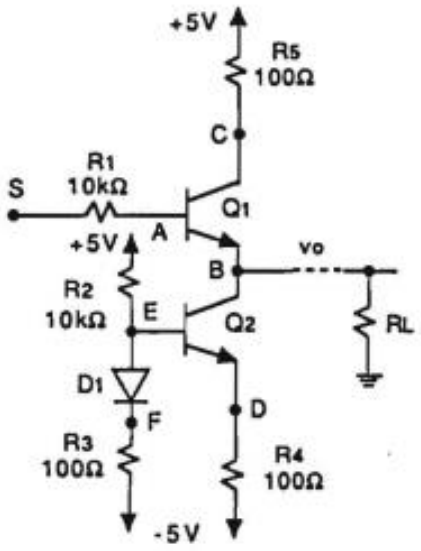


Figure 2: Class-A BJT Follower with Emitter-Current Bias

a) Assemble the circuit as shown in Fig. 2 with S grounded and no load connected to node B. Adjust the supplies to ±5 V as closely as you can. Measure the voltages al nodes A, B, C, D, E, F. Estimate the collector current of Q_1 and its ß (Do not forget measuring resistance values of $R_1$and R_5).

b) Shunt R_2 by a 1 kΩ resistor, and remeasure all nodes. What is the bias current in Q_1? What is its ß at this current level?

Hint: Consider ß and r_e at the current levels you have measured. Notice that the currents in $D_1$and Q_2 are nearly the same (due to R_3, R_4).

Table 1: DC values of Class-A amplifier

	V_A	V_B	V_C	V_D	V_E	V_F	I_C	I_B	β
a)	…	…	…	…	…	…	…	…	…
b)	…	…	…	…	…	…	…	…	…

1.2 Signal Operation:

Connect the circuit as shown in Fig. 2 with R_2 = 10kΩ, a load R_L = 10kΩ, and node S connected to a generator providing a 0.1 Vpp triangle wave at 1kHz. With your oscilloscope, measure the peak to peak voltages at nodes S, A, B. Calculate the voltage gains from S to B and A to B, as well as the input resistance at A.

Table 2: AC values of Class-A Amplifier

	v_s	v_a	v_b	v_b/v_s	v_b/v_a	R_{in}
R_L = 10kΩ	…	…	…	…	…	…
R_L = 1kΩ	…	…	…	…	…	…

While observing nodes S and B, with both oscilloscope channels direct-coupled with zero volts at the screen center, raise the input voltage until first one output peak, and then the other, limits. At what peak voltages does limiting occur?

Reduce the load (R_L) to 1kΩ and apply, initially, a 0.1 Vpp triangle wave at 1kHz. While observing nodes S and B, raise the input amplitude until first one peak and then the other peak limit. Observe the critical peak voltages at both input and output. Note the input voltages (v_s) when output voltage reaches at positive peak limit (v_B^+) and negative peak limit (v_B^-).

Table 3: Critical Peak Voltages of Class-A amplifier

	v_s when v_B^+	v_s when v_B^-
R_L = 10kΩ	…	…
R_L = 1kΩ	…	…

Hint: Consider the new voltage gain, input resistance, and the nature of the peak output-voltage limitation. Note the relationship between the class-A bias current and the available output swing in one direction!

2. Class B Follower

2.1 DC Bias:

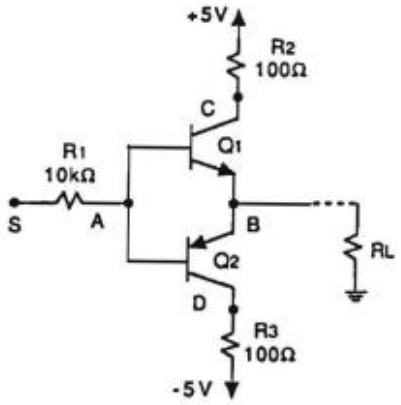


Figure 3: Class-B Complementary BJT Output Stage

Assemble the circuit in Fig. 3 with node S grounded, R_L = 10 kΩ, and supplies at ±5V. Measure the voltages at nodes A, B, C, D.

Table 4: DC values of Class-B amplifier

V_A	V_B	V_C	V_D
…	…	…	…

2.2 Signal Operation:

With a 0.2 Vpp triangle wave applied to node S initially, and a 10 kΩ load connected to node B, measure nodes S, A and B. Now, observing node B, increase the input signal until the output peaks just begin to limit. Note their value and the corresponding peak values at nodes S and A.

Table 5: AC values of Class-B Amplifier

v_s	v_a	v_B	v_S^+	v_A^+	v_B^+	v_S^-	v_A^-	v_B^-
…	…	…	…	…	…	…	…	…

Hint: Note the non-ideal performance for small signals (and the open circuit behaviour for very very small ones). Note the separate roles of the output transistors depending on output polarity and load-current direction.

3. Diode Biased Class – AB Output Stage

3.1 DC Operation:

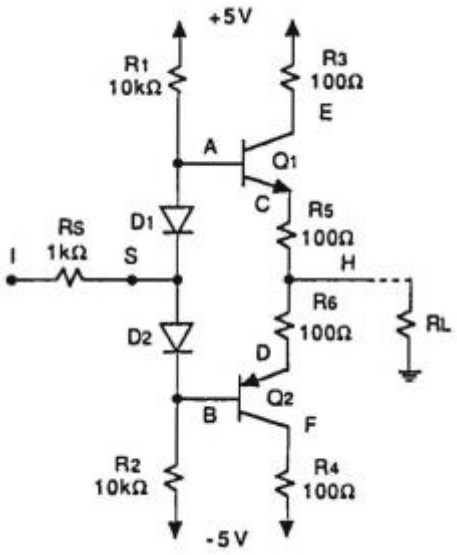


Figure 4: Class-AB Follower

Assemble the circuit in Fig. 9.4 with node I grounded, no load connected, and supplies adjusted to ±5.0 V. Measure nodes S, A, B, C, D, E, F, H. Estimate the bias current in $Q_1$and Q_2.

Table 6: DC values of Class-AB Amplifier

V_S	V_A	V_B	V_C	V_D	V_E	V_F	V_H	I_1	I_2
…	…	…	…	…	…	…	…	…	…

3.2 Signal Operation:

To the circuit of Fig. 4, connect a triangle-wave generator to node I initially set to 0.2 Vpp at 1kHz, and connect a 10 kΩ load to node H.Measure the voltages at nodes S, I and H. Then, raise the input voltage to find the largest unclipped output available. Measure the peak voltages at I, H.

Table 7: AC values of Class-AB Amplifier

v_s	v_i	v_h	v_i^+	v_h^+	v_i^-	v_h^-
…	…	…	…	…	…	…

Hint: Consider the nature of output-signal limitation, in particular the roles of collector voltage, current-equalizing and control components, and device ß.
Tip for Analysis and Discussion: Compare all the different output stages we have examined in this experiment. Point out their characteristics and the differences between them.

REFERENCES:

Smith, K. C. and Sedra, A. S., *Laboratory Explorations for Microelectronic Cirtuits*, Oxford University Press, 1998.

NOTES